

NEURO-SYMBOLIC LEGAL INVARIANT CHECKER

Formal Verification of Financial Credit Agreements • Complete Plain-English Guide

1. The Simple Story: What Problem Are We Solving?

The Real-World Situation:

Imagine a massive company wants to borrow **₹500 Crores** from a major bank. Their corporate legal teams draft a **120-page legal contract** filled with hundreds of financial rules and covenants.

Because different legal teams draft different clauses late at night, **hidden contradictions slip into the contract**:

- **On Page 20 (Section 6.01)**: The contract states: *'If the company acquires a competitor, they are permitted to increase their debt ceiling up to 4.0x their earnings.'*
- **On Page 85 (Section 6.02)**: Another clause states: *'In no event shall total debt exceed ₹30 Crores whenever earnings drop below ₹8 Crores.'*

The Disaster: What Happens When Both Clauses Collide?

Suppose the company earns **₹7.75 Crores** and borrows **₹31 Crores** to acquire a supplier.

- According to **Page 20**: It is **ALLOWED** (Leverage Ratio = $31 / 7.75 = 4.0x$ threshold).
- According to **Page 85**: It is **ILLEGAL** (Debt ₹31 Cr > ₹30 Cr while earnings ₹7.75 Cr < ₹8 Cr).

Result: The contract is mathematically contradictory! In practice, this ambiguity triggers lawsuits, loan recall defaults, and multi-crore legal disputes.

Why Can't We Just Use ChatGPT or Claude?

1. **ChatGPT Guesswork & Hallucinations**: Large Language Models predict text probabilistically. While excellent at summarizing essays, *they cannot perform rigorous formal mathematical proofs.*
2. **Missed Boundary Edge Cases**: Standard LLMs will read Section 6.01 and 6.02 and declare: *'This agreement looks standard and sound.'* They cannot evaluate millions of numerical permutations simultaneously.

2. What YOUR Application Does: The 3-Step Pipeline

Our system is a **Neuro-Symbolic Legal Invariant Checker**. It bridges the intuitive linguistic understanding of modern AI with the mathematical certainty of Microsoft Z3 SMT logic solvers:

Phase 1: Neural Semantic Parsing

Gemini 2.5 Flash API

Acts as a legal translator. Reads raw unstructured text from PDFs and parses covenants into a strict Abstract Syntax Tree (AST) schema.

Phase 2: Symbolic Verification

Z3 SMT Solver (Microsoft)

Translates AST into first-order logic on CPU. Formulates mathematical invariant refutation. Checks all edge cases in under 50 milliseconds.

Phase 3: Legal Redlining

Automated Reconciliation

Extracts counterexample numerical values (\$31M Debt, \$7.75M EBITDA) and generates legal redline recommendations.

3. Dictionary of Difficult Terms (Plain English for Viva)

Memorize these simple real-world analogies to answer any professor questions with ease:

Complex Term	What It Sounds Like	What It ACTUALLY Means (Simple Analogy)
Neuro-Symbolic	Sci-Fi Alien brain	The Brain + The Calculator : Combining a neural network (Gemini) to read language with a symbolic engine (Z3) to do pure math logic.
Invariant	Deep mathematical rule	A rule that must NEVER be broken : Like a 'Never exceed 80 km/h' speed limit. In our app: 'Debt ratio must never exceed 3.5x'.
SMT Solver (Z3)	Abstract research tool	A hyper-fast logic calculator built by Microsoft : You give it 50 rules and it verifies whether all rules can hold simultaneously without contradiction.
AST (Abstract Syntax Tree)	Complex data structure	The mathematical skeleton : Converts paragraphs of legal text into clean variables like $\text{LeverageRatio} = \text{Debt} / \text{EBITDA}$.

Counterexample	Theoretical proof item	The proof of guilt: When Z3 finds a flaw, it prints the exact numbers: 'At Debt = \$31M and EBITDA = \$7.75M, the contract breaks!'
Refutation Principle	Formal logic method	Proving safety by searching for a bug: We ask Z3: 'Can you find ANY scenario where rules break?' If Z3 says NO (UNSAT), the contract is proven safe.
Covenant	Medieval legal oath	A binding promise in a loan: e.g., 'The borrower promises to keep at least █10 Crores in their bank account at all times.'
EBITDA	Wall Street buzzword	Operational Cash Profit: Earnings Before Interest, Taxes, Depreciation & Amortization. Basically: how much cash the core business generates.
Leverage Ratio	Financial metric	Debt divided by Profit (Debt / EBITDA): If a company owes █30 Cr and earns █10 Cr, their leverage ratio is 3.0x.
Redlining	Document editing	Automated clause fix: Marking flawed legal text and inserting clean conflict-override phrases (e.g. 'Subject to Section 6.02').

█ 4. The 1-Minute Pitch (Memorize This for Your Viva!)

"Sir, large financial credit agreements span hundreds of pages and contain complex financial formulas. Human lawyers frequently make drafting mistakes where two separate clauses contradict each other.

Large Language Models like ChatGPT cannot reliably detect these issues because they hallucinate and cannot perform formal mathematical proofs.

Our project implements a **Neuro-Symbolic architecture**:

1. **Gemini AI** extracts unstructured contract covenants into a formal mathematical syntax tree.
2. Microsoft's **Z3 SMT solver** executes mathematical refutation logic on CPU in milliseconds to prove whether invariants can ever be broken.
3. When a conflict exists, our system pinpoints the exact numerical counterexample and generates automated legal redlines to fix the contract before signing."